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Energy-Aware Data Transfer Tuning

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***Abstract*—The annual electricity consumed by data transfers in the U.S. is estimated to be 20 Terawatt hours, which translates to around 4 billion U.S. Dollars per year. There has been considerable amount of prior work looking at power management and energy efficiency in hardware and software systems, and more recently in power-aware networking. Despite the growing body of research in power management techniques for the networking infrastructure, there has been no prior work (to the best of our knowledge), focusing on saving energy at the end-systems (sender and receiver nodes) during the data transfer. We argue that although network-only approaches are part of the solution, the end-system power management is a key in optimizing energy efficiency of the data transfers, which has been long ignored. In this paper, we analyze various factors that will affect the power consumption in end-to-end data transfers, such as the level of parallelism, concurrency and pipelining. Our results show that significant amount of energy savings can be achieved at the end-systems during data transfer with no or minimal performance penalty.**

***Keywords*-Power-aware data transfers; Energy efficiency; Big-data; Protocol tuning; Power modeling;**

I. Introduction

The global data movement requirements have already exceeded the Exabyte scale. According to the Global Internet Traffic statistics by Cisco [1], a total of 376 Exabytes of data was moved from one location to another in year 2012. U.S. generated 49 Exabytes of this global data traffic. The annual electricity consumed by these data transfers in the U.S. is estimated to be 20 Terawatt hours, which translates to around 4 billion U.S. Dollars per year [2]. One quarter of this electricity is consumed by the end-systems (sender and receiver nodes) and the rest by the networking infrastructure. There has been considerable amount of prior work looking at power management and energy efficiency in hardware and software systems [3]–[13] and more recently in power-aware networking [14]–[18].

Almost all existing work on power-aware networking focuses on reducing the power consumption on networking devices (i.e. switches, routers etc.) Some of the proposed techniques in this area include putting idle sub-components (i.e. line cards, etc.) to sleep [19]; adapting the rate at which switches forward packets depending on the traffic [20]; developing new architectures with programmable switches [21] and switching layers that can incorporate different policies [22]; and power-aware network protocols for energy-efficiency in network design and routing [23].

Despite the growing body of research in power management techniques for the networking infrastructure, there has been no prior work (to the best of our knowledge) focusing on saving data transfer energy at the end-systems (sender and receiver nodes). We believe that significant amount of data transfer energy savings can be obtained at the end-systems with no or minimal performance penalty. We argue that although network-only approaches are part of the solution, the end-system power management is a key in optimizing energy efficiency of the data transfers, which has been long ignored. While achieving energy efficiency, we should avoid panelizing the performance, since there is substantial empirical evidence suggesting that performance directly impacts revenue. As two well known examples for this, Google reported 20% revenue loss due to a specific experiment that increased the time to display search results by as little as 500 milliseconds; and Amazon reported a 1% sales decrease for an additional delay of as little as 100 milliseconds [24].

In this paper, we analyze various factors that affect the power consumption in end-to-end data transfers. Different protocol parameters such as TCP pipelining, parallelism and concurrency levels play a significant role in the achievable network throughput. However, setting the optimal numbers for these parameters is a challenging problem, since not-well-tuned parameters can either cause underutilization of the network, may increase the power-consumption drastically, or may overload the network and degrade the performance due to increased packet loss ratio, end-system overhead, and other factors. Our results show that significant amount of energy savings can be achieved at the end-systems during data transfer with no or minimal performance penalty.

The rest of the paper is organized as follows. Section 2 describes our power model for data transfers and how we achieve energy efficiency in end-to-end data transfers by minimizing the power consumption at the end-systems. Section 3 presents our experimental results, section 4 describes the related work, and section 5 concludes the paper.

II. Power Modeling for Data Transfers

Power models have crucial importance in energy efficiency studies due to lack of hardware power monitoring devices on large-scale systems. Power management systems such as on-board power sensors and active energy managers recently became built-in features in state-of-the-art servers [25] but most legacy systems still do not have such features. Even though one can measure power consumption of a server, an application level power modeling is still necessary to calculate power consumption of each application from the total power

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System	Intel Server	AMD Server
CPU	3.07GHz Intel Xeon	2.8GHz AMD FX
Memory	6GB DDR3	10GB DDR3
Storage	500GB 3.5" Hard disk	2TB 3.5" Hard disk
Network	1 Gigabit Ethernet	1 Gigabit Ethernet
OS Kernel	Linux 3.2.0	Linux 3.2.0

TABLE I: Specifications of servers used in experiments

consumption. Thus we have developed two models to analyze power consumption for two types of access privileges. The fine-grained power model requires access to utilization of all system components, on the other hand, CPU-based power model only needs utilization of the CPU in the system.

Our approach resembles Mantis [8] in predicting power consumption of data transfers based on operating system metrics. This approach requires a onetime model building phase to extract power consumption characteristics of the system components. For each system component, we measure power consumption values for varying load levels. Then, linear regression technique is used to derive the coefficients for the component metric. The derived coefficients are used in the power model to predict power consumption of the data transfers.

A. Modeling Architecture

Our power modeling architecture has five main components: i) source server; ii) destination server; iii) power meter; iv) transfer controller; and v) power estimator.

The *source and destination servers* are the end-points which we run data transfer in between. They are dedicated for high-performance data transfers. We have used two different types of servers in our experiments. The configuration of these servers are listed in Table I.

The *power meter* is used to measure the real power consumption of the data transfer servers. The power consumption values are used to derive our models in the model building phase, and to validate the accuracy of our power model in the evaluation phase. In our experiments, we use highly accurate and precise Yokogawa WT210 power meter. It is one of the accepted devices by SPEC power community for power analysis studies.

The *transfer controller* initiates and finalizes data transfers between the source and destination servers. The *power estimator* is used in the model building phase, where it collects and synchronizes the utilization information from source/destination servers and the power consumption values from power meter. Then, it derives coefficients of each system component and builds the power models which are later used in experiments to estimate the power consumption of data transfers.

B. Model Building Phase

In the model building phase, we performed transfers of different sized data sets with different transfer tools (i.e., scp, rsync, ftp, gridftp, bbcp) to analyze the power consumption behavior of data transfers. In these transfers, we collected utilization and I/O request measurements of main components at the system and process levels using the *utilization collector tool*. The utilization collector tool is a custom implemented software which collects utilization and I/O request values per second using bunch of low-level system tools (i.e., perf, sar, pidstat) and then sends this data to the transfer controller and the power estimator unit.

The system components with most impact to the data transfer power consumption are CPU, memory, hard disk, and network interface card (NIC). We used Gamut [26] and lookbusy [27] tools to extract power consumption characteristics of these components. Gamut and lookbusy are used to stress each component individually at various levels of utilization. Using these utilization values, the power estimator unit derives coefficients for each component using linear regression. The power values used in regression are obtained by subtracting idle system power from total system power. Idle system power is the power consumption of server when only operating system processes are running and utilization of all system components is minimized. The derived model is then used to predict energy consumption of data transfer tools.

CPU is the most significant component in power consumption of data transfer servers. We observed that the power consumption behavior of multicore-processors is different than that of single core processors. Power consumption of multi-core processors is not simply equal to sum of each individual core's power consumption. To analyze this behavior we used lookbusy that can generate synthetic workload on CPU cores at certain utilization rate. On our Intel server, we created synthetic workload on different number of cores by increasing utilization 5% at each step. As shown in Figure 1, the number of active cores affects power consumption significantly, and per core power consumption is reverse proportional to the number of active cores. For example, when 1 core is loaded with 100% utilization its power consumption becomes 32.2 watts. However, when all 4-cores are loaded with 100% utilization, total power consumption increases to 85.6 watts which reduces per core power consumption to 21.4 watts. Thus, we can conclude that instead of running one core at high utilization rate, it is power efficient to distribute load on multiple cores at less utilization rate. In our model, we took number of active cores used by transfer tool into account when calculating power consumption of transfer tool.

C. Fine-Grained Power Model

The four operating system metrics used in our fine-grained model were:

- u_{cpu}: CPU utilization
- u_{memory} : Memory access rate
- u_{disk}: Hard disk I/O request rate
- u_{net}: Network I/O request rate

The equation form of fine-grained power model at time t is as follows:

$$\begin{aligned} P_t = & C_{cpu,n} \times u_{cpu,t} + C_{memory} \times u_{memory,t} + \\ & C_{disk} \times u_{disk,t} + C_{nic} \times u_{nic,t} \end{aligned} \quad (1)$$

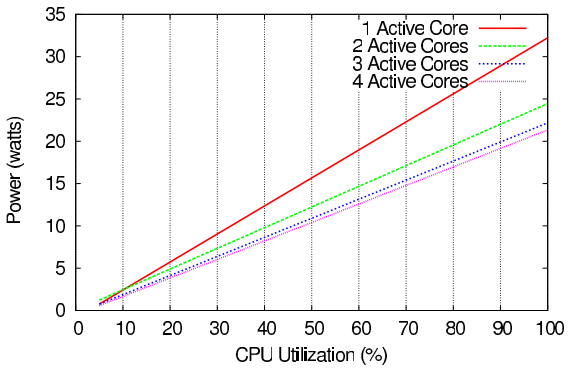


Fig. 1: Power consumption behavior of data transfer server with intel multi-core processor

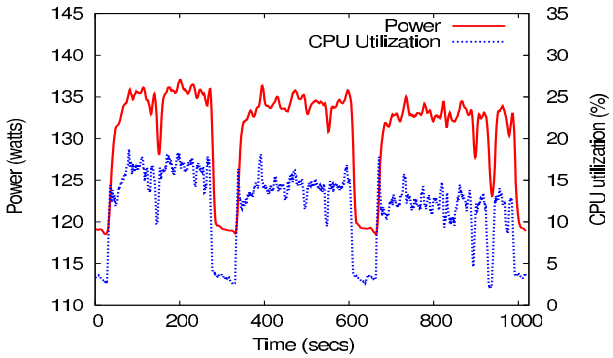


Fig. 2: Stacked graph of CPU utilization versus power consumption during data transfers.

$$C_{cpu,n} = 0.016 \times n^2 - 0.112 \times n + 0.414 \quad (2)$$

P_t refers to predicted power consumption of data transfer tool at time C_{cpu} , C_{memory}, C_{disk} and C_{nic} are the coefficients of CPU, memory, disk and NIC respectively. Since we used multi-core processor servers, coefficient of CPU depends on number of active cores in the transfer as given in Equation 2.

D. CPU-based Power Model

In typical data centers, hardware power monitoring option is not available for servers or not accessible by end users. Furthermore, in these environments accessing disk, memory and network utilization of a process is also restricted for end-users. Previous studies showed that solely CPU-based power prediction models can perform slightly worse compared to fine-grained models. As shown in Figure 2, CPU utilization and power consumption expose highly related behaviors during data transfer. Correlation between CPU and consumed power is 89.71%. Hence, we assumed that CPU-based models can give us yet accurate results for our experiments, thus lets us to measure energy consumption of data transfer tools in data center environment.

We have developed a new extendable power model with the help of nameplate power value of CPU component. We used CPU Thermal Design Power (TDP) value to scale power drawn of data transfer tools at different servers. TDP values are provided by CPU vendors that refers to the maximum power a processor can draw for a thermally significant period while running commercially useful software. The main contribution of this model is, it allows to predict power consumption of a server by just using CPU utilization values and calculating a machine-specific CPU coefficient by taking ratio of TDP values.

The equation form of CPU-based power model at time t is as follows:

$$P_t = (C_{cpu,n} \times u_{cpu,t}) \times \frac{TDP_{SR}}{TDP_{SL}} \quad (3)$$

where SL is the server we used to define CPU-based model and SR is a server we want to extend the CPU-based model. TDP_{SR} is the TDP value of the server SR and TDP_{SL} is the TDP value of the server SL. $u_{cpu,t}$ is utilization of server SR for a data transfer.

E. Evaluation of Power Models

We have used Intel and AMD servers to evaluate fine-grained and CPU-based power models . The configurations of these servers are listed in Table I. These servers have different processor types from different vendors. Both servers have similar memory and hard disk properties. We implemented the power models on the Intel server.

We have used five different transfer tools to evaluate proposed models. These tools are;

- scp: standard secure copy tool
- rsync: standard directory synchronization tool
- ftp: standard file transfer protocol tool
- bbcp: peer-to-peer file transfer tool
- gridftp: high-performance and secure data transfer tool based on ftp

We generated dataset of size 20GB, consisting of various sizes of files to transfer the data between servers. We evaluated the models by comparing power prediction values with power meter values. We subtracted the idle system power from total power when using power meter values. The transfers are remotely started by transfer controller unit. We repeated the transfers ten times for each tool and the errors from each run are aggregated.

We tested the fine-grained model on Intel server and CPU-based model on both Intel and AMD servers. Figure 3 shows the mean absolute percentage error rate of each tool for each model. Mean absolute error rate of a model is calculated using

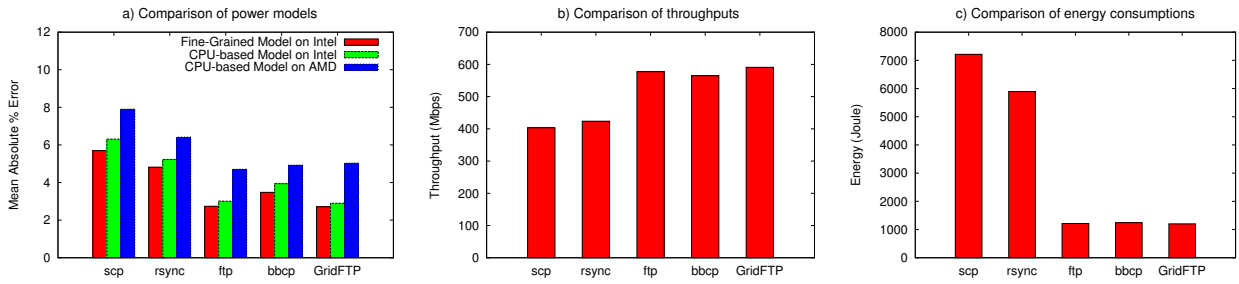


Fig. 3: Analysis of data transfer tools.

the following equation:

$$error = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{P_{pred} - P_{act}}{P_{act}}\right| \times 100 \quad (4)$$

where P_{pred} is the power predicted by model, and P_{act} is the actual power measurement.

Figure 3 (a) shows the error rates of the models for different data transfer tools. The fine-grained model has the lowest error rate for all tools as expected. Its error rate is always below 6%. The error rate of CPU based-model for Intel server is very close to that fine-grained model. This proves that CPU-based model performs as good as the fine-grained model when it is tested on the server where it is implemented. That is, we used Intel server to develop CPU-based model so that it performs similar to fine-grained model when tested on the Intel server. On the other hand, when we extend the CPU-based model to AMD server by proportioning its TDP value to Intel server's TDP value, the error rate increases 2-3% compared to fine-grained model. Even though the error rate of CPU-based model increases, it is still less than 8% for all tools and less than 5% for ftp, bbcp and gridftp.

The throughput and power consumption of each transfer tool are given in Figure 3 (b) and (c). Even though scp and rsync are the popular transfer tools, they perform the lowest throughput. Ftp, bbcp, and gridftp perform almost 1.5 times better compared to scp and rsync in terms of data transfer throughput. Regarding to power consumption, scp, and rsync consumes 5-6 times more energy compared to other tools. This shows that scp and rsync does not only yield less throughput but also causes much more load on the CPU compared to other tools.

F. Applying the Model

We applied our power model to analyze the effect of three transfer parameters on end-to-end throughput as well as power consumption. These parameters are TCP pipelining, parallelism and concurrency levels.

Pipelining specifically targets the problem of transferring large numbers of small files [28], [29]. In a regular TCP transfer, acknowledgement for the previous transfer is waited before starting the next transfer, which may cause a delay of more than one RTT between individual transfers. With pipelining, multiple unacknowledged transfers can be active in the network pipe at any time and the delay between individual transfers is minimized. As an example, if the pipelining level is three, the client can issue three outstanding transfer commands and the transfers are lined up back-to-back in the same data channel. Whenever a transfer finishes, a new command is issued to keep the pipelining queue full. Pipelining prevents the data channel idleness and eliminates the idle time due to control channel conversations in between the consecutive transfers. It also prevents TCP window size from shrinking to zero due to idle data channel time if it is more than one Round Trip Time (RTT).

Parallelism sends different chunks of the same file using different data channels (TCP streams) at the same time, and achieves high throughput by mimicking the behavior of individual streams and getting an unfair share of the available bandwidth [30]–[36]. On the other hand, using too many simultaneous connections reaches the network on a congestion point and after that threshold the achievable throughput starts to drop down. Predicting the optimal parallel stream number for a specific setting is a very challenging problem due to the dynamic nature of the interfering traffic on the intermediate links. Hacker et al claimed that the total number of streams behaves like one giant stream that transfers in capacity of total of each streams' achievable throughput [37]. However, this model only works for uncongested networks, since it accepts that packet loss ratio is stable and does not increase as the number of streams increases. Another study [38] declares the same theory but develops a protocol which at the same time provides fairness. Dinda et al. modeled the bandwidth of multiple streams as a partial second order polynomial equation which needs two different throughput measurements of two different stream numbers to predict the others [39].

Concurrency refers to sending multiple files simultaneously through the network using different data channels at the same time. Most studies in this area do not take the data size and the network characteristics into consideration when setting the concurrency level [40], [41]. Liu at al. [42] adapt the concurrency level based on the changes in the network traffic, but do not take into account the other bottlenecks that can occur on the end systems.

Specs	Local Intel-Local AMD	Local Intel-SDSC	Stampede-SDSC
Bandwidth	1 Gbps	1Gbps	10 Gbps
RTT	0.2 ms	80 ms	60 ms
TCP Buffer Size	32 MB	32 MB	32 MB
BDP	0.02 MB	10 MB	72 MB
File System	NFS-NFS	NFS-Lustre	Lustre-Lustre

TABLE II: Network specifications of test environments

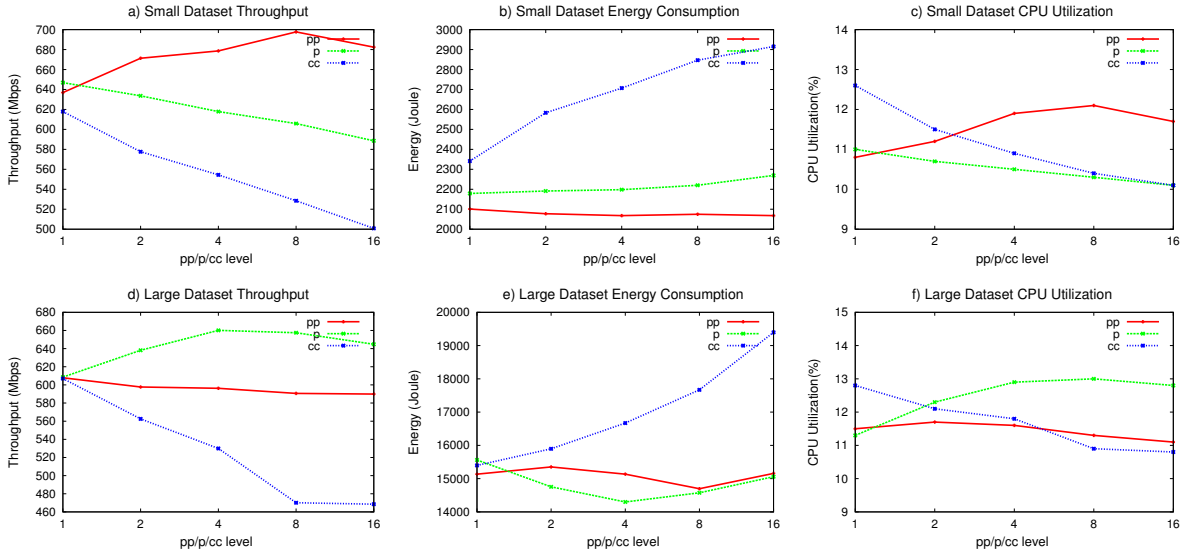


Fig. 4: Local Area Data Transfers

III. Experimental Results

We used our model to analyze the power consumption during GridFTP data transfers with different levels of concurrency (cc), parallelism (p), and pipelining (pp). We used two different datasets during these experiments: a small dataset compromised of files in varying sizes between 5MB-10MB with a total size of 20GB; and a large dataset containing files with 100GB total size and file sizes ranging between 10GB-20GB.

Figure 4 presents the results of data transfer between two servers in LAN. In Figure 4(a), we can see that the only parameter that improves the throughput of the small dataset is pipelining. On the other hand, parallelism and concurrency lead to lower transfer throughput as we increase their level. For the large dataset, shown in Figure 4(d), parallelism increases the throughput by 8% when its value is increased from 1 to 4. However, when parallelism is set to a level higher than 4, then the throughput decreases an in turn triggers an increase in power consumption. This proves that parallelism level can improve the throughput and decrease the power consumption of large-file dominated dataset transfers if it is set to the correct value. Otherwise, it might cause lower throughput and higher power consumption. For both datasets, concurrency has undesirable effect as it decreases the throughput and increases power consumption. It increases the power consumption by 22-24% as its level increases from 1 to 16. The main reason behind this behavior is having a single disk storage subsystem in the local servers which performs worse as the number of synchronous disk accesses increase.

Figure 5 presents the results of data transfer between a server at DIDCLab in Buffalo, New York and an XSEDE node at SDSC in San Diego, California. The roundtrip time is around 80 ms. Since there is no dedicated network between the source and destination servers, and a shared internet connection is utilized, it is hard to say about maximum attainable network bandwidth. However, our results showed that we can achieve up to 150 Mbps throughput.

The GridFTP protocol parameters affect in a different manner when destination disk subsystem is changed to the parallel filesystem (Lustre) which shows that transfer bottleneck was disk write capacity in LAN tests. In cases where opening a new transfer channel (either by increasing the parallelism or concurrency levels) increases the transfer throughput considerably (as shown in Figures 5(a) and 5(d)), the imposed CPU overhead of channel management becomes insignificant compared to CPU usage increase due to handling received data. In this low latency, low bandwidth network, concurrency is the parameter that improves transfer throughput the most (Figure 5(a)). As we increase the concurrency level, throughput jumps significantly from 15 Mbps to 100 Mbps for the small dataset (567% increase), and from 70 to 110 (57% increase) for the large dataset . Although CPU utilization also increases,

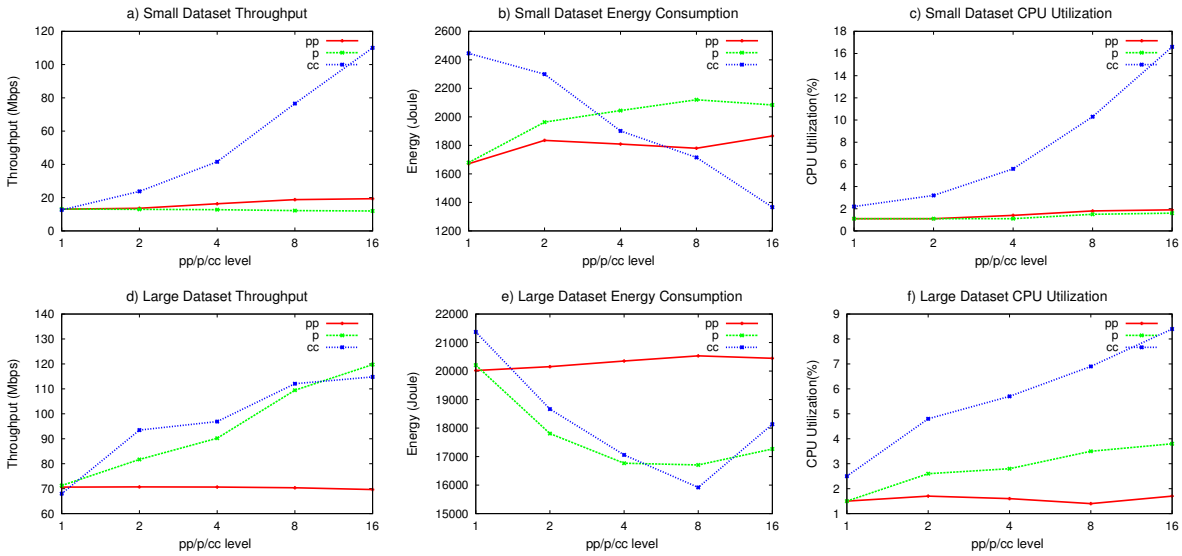


Fig. 5: Transfer from local server to XSEDE node

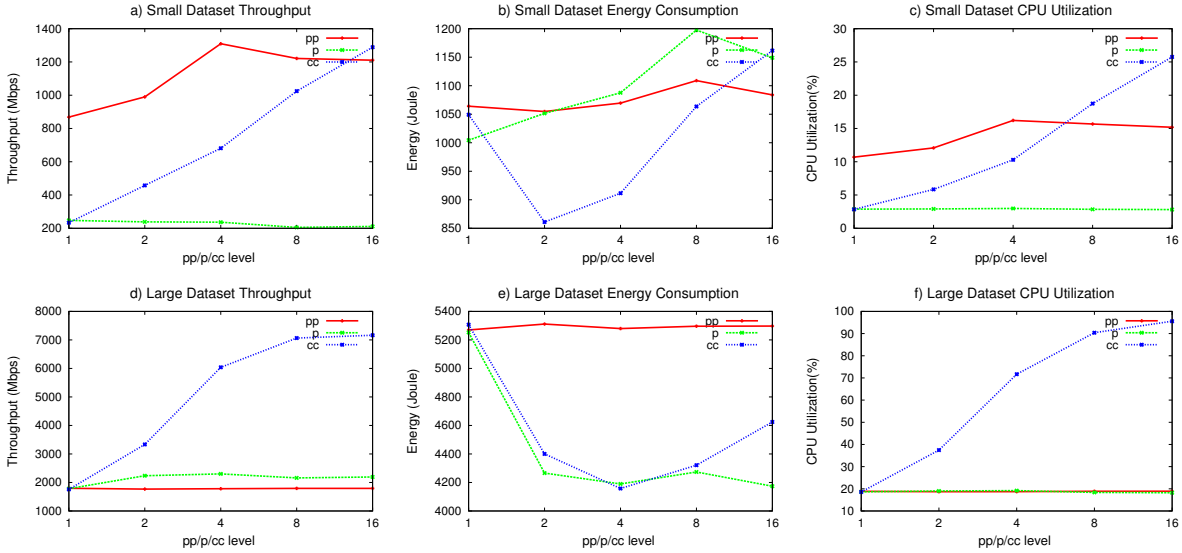


Fig. 6: Transfer between two XSEDE nodes

the overall power consumption decreases significantly. The reason behind this derives from leveraging from all cores in the CPU. As we discussed in Section II.A, as number of active cores increases the coefficient $C_{cpu,n}$ in CPU-based model (Equation 3) decreases which is an expected behavior. We found $C_{cpu,n}$ becomes 0.231 when single core is used and it decreases to 0.156 when all four cores are used. In Figure 5(a), we observe that increased level of parallelism does not affect the transfer throughput for small files while CPU utilization increases little bit as given in Figure 5(c). This leads increase in total power consumption as in Figure 5 (b). For large files, parallelism increases the throughput by around 70%, while decreasing the power consumption by 16%.

Figure 6 shows the effect of transfer protocol parameters in wide area high speed network transfers. These experiments were run between two XSEDE nodes, with 10 Gbps bandwidth and 60 ms round trip time. Both nodes had Lustre file system. In this setting, pipelining increases the transfer throughput around 40% for the small dataset when parameter value is increased from 1 to 16 as shown in Figure 6(a). This increase in throughput increases the CPU utilization by around 37%, and the overall power consumption reduces around 5%. Although the gain in power consumption seems low, it comes with no cost thus use of pipelining is definitely helpful for

both to increase speed of the transfer and reduce power consumption. For the large dataset, pipelining has almost no effect. Concurrency increases the throughput significantly for both datasets. Power consumption is reduced by 20% when concurrency level is set to 4 as compared to when it is set to 1 for large files. Further increase in concurrency, however, lowers power consumption ratio to 9% when it is set to 16. For small files, concurrency level 2 yields lowest power consumption which is different than the large file transfer case. Hence, we can say that although increasing concurrency increases throughput at all levels, the slope of throughput curve reduces after some level as can be seen in Figures 6(a) and (d). Since CPU utilization curve does not follow the same pattern with throughput curve as varying level of concurrency, it leads to an increase in total power consumption of the data transfer. So, one can set the level of concurrency based on the objectives. For example, if the objective is to obtain maximum transfer throughput, concurrency level 16 will be the choice for large dataset. On the other hand, the concurrency level 4 will be chosen if the objective is to minimize the power consumption with a reasonable throughput.

IV. Related Work

To the best of our knowledge, there has been no prior work focusing on saving energy at the end-systems (sender and receiver nodes) during the data transfer. On the other hand, there has been numerous studies analyzing and predicting the power consumption of servers and data centers using power models. We can categorize these studies into three broad classes: *(i)* simulation based, *(ii)* hardware and software performance metrics based, and *(iii)* application based. The initial works in this area are simulation-based approaches which focused on predicting power consumption of various hardware component such as CPU [3], memory [4], disk [5], and network [43]. There are studies to predict system-level power consumption based on simulations such as SoftWatt [6]. These models are developed for off-line prediction so, they are limited in terms of portability, speed and accuracy.

Contreras et al. [7] used hardware event counters to predict power estimation at the system level. However, hardware event counters have different characteristics for different processors and it is difficult to reach them in the restricted shared environments. In Mantis [8], server level power consumption is predicted by means of OS-level utilization and counter metrics. Their approach predicts the power consumption with low error rates, which makes it very popular in the area.

Fan et al. [9] only uses CPU utilization information to build prediction model, and Heath et al. [44] uses CPU and disk utilization information as main input parameters in their model. These works performed similar prediction accuracy as in Mantis [10]. In our study, we took Mantis as a starting point of our model then extended it to predict power consumption of various data transfer applications.

The number of application level power analysis studies are relatively less than system-level studies. Most of these works are fall into energy profiling area category [11], [45], [46]; however,they require a long term profiling steps yet they are limited in terms of accuracy and feasibility. Another process level study is pTop [47] which is a process-level power prediction tool based on nameplate power consumption values of hardware components. However, relying only on nameplate values of each component may lead inaccurate prediction method. Fan et al. [9] proved that the actual power consumption of the server is less than nameplate values in most cases.

Most of the studies in the area of energy efficiency in data centers focus on workload consolidation in order to reduce the number of the active servers [48]–[51]. Susan et al. [13] offers smart replication method as a way of power consumption reduction. They proposed a new power efficient replication algorithm to maximize CPU utilization to minimize idle power usage similar to workload consolidation techniques . Harnik et al. [52] focused on power consumption of disk storage systems. They analyzed how system performance get affected when disks are working on a low-power mode. Hasebe et al. [12] used virtual nodes and migration mechanisms to minimize active disk numbers. They also proposed a new replication algorithm to minimize data the migration costs.

The work on power-aware networking focuses on reducing the power consumption on networking devices (i.e. switches, routers etc.). Gupta et al. [53] were amongst the earliest researchers to advocate conserving energy in networks. They suggested different techniques such as putting idle subcomponents (i.e. line cards, etc.) to sleep [19], which were later extended by other researchers. S. Nedevshi et al. proposed adapting the rate at which switches forward packets depending on the traffic [20]. Other related research in power-aware networking has focused on architectures with programmable switches [21] and switching layers that can incorporate different policies [22]. Barford et al proposed power-aware network protocols for energy-efficiency in network design and routing [23]. We argue that although network-only approaches are part of the solution, the end-system power management is a key in optimizing energy efficiency of the data transfers, which has been long ignored by the community.

V. Conclusion

We analyzed different parameters that affect both the throughput and the power consumption in end-to-end data transfers. Different protocol parameters such as TCP pipelining, parallelism and concurrency levels play a significant role in the achievable network throughput. However, setting the optimal numbers for these parameters is a challenging problem, since not- well-tuned parameters can either cause underutilization of the network, may increase the power-consumption drastically, or may overload the network and degrade the performance due to increased packet loss ratio, end-system overhead, and other factors. Our model can capture how these parameters affect the throughput and power consumption.

Our experimental results show that significant amount of energy savings can be achieved at the end-systems during data

transfers with no or minimal performance penalty. In some cases, both the throughput can be maximized and the power consumption can be minimized at the same time, which is a win-win situation. In our future work, we are planning to analyze the joint effect of different parameters combinations both on performance and on power consumption. The ability to select the ideal set of transfer parameters for high performance, low cost (in terms of energy) data transfers will be invaluable for many applications that require large data movements.

ACKNOWLEDGMENT

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